Cost-Effective Predictive Modeling for Customer Targeting

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Abstract

We address profit-oriented customer targeting on a binary dataset of 5 000 observations and 500 anonymous features. The business score $10\,\mathrm{TP} - 5\,\mathrm{FP} - 200\,N_{\mathrm{vars}}$ couples predictive performance with model complexity, penalising both false positives and the number of selected features. We compare a classic cross-validation pipeline—exploring sequential feature selection and five modelling strategies—with a nested cross-validation alternative that evaluates over 257 prescreening strategies and 42 model configurations. The nested approach improves the pooled out-of-fold business score from 3 380 to 5 425, a 61% gain, using six features selected by a sparse GAM prescreener paired with a shallow ExtraTrees classifier.

Index Terms: feature selection, profit maximization, customer targeting, ensemble methods

1. Introduction

Cost-aware customer targeting optimises a profit function rather than a standard classification metric. When each additional declared feature carries a direct penalty, the feature set size becomes a first-class model parameter, and discriminative power alone no longer determines the optimal setup. This creates a setting where compact, well-calibrated rankers should outperform richer models that would dominate on ROC-AUC or F1.

We study this problem on a 5 000-observation binary dataset with 500 anonymous features and a score

$$\mathrm{Score} = 10\,\mathrm{TP} - 5\,\mathrm{FP} - 200\,N_{\mathrm{vars}},$$

where each extra variable requires 20 additional true positives to break even. Two approaches are developed. Section 2 describes the initial pipeline: sequential dimensionality reduction and cross-validated comparison of five modelling strategies (Section 2.2–2.8). Section 3 presents the nested-CV alternative (Section 3), which refits all prescreening estimators inside each outer fold and runs a two-round grid search over the joint feature–model space. Results are compared in Section 4.

2. Methodology

This section describes the base pipeline: a conventional cross-validation workflow used as the primary modelling framework and as a performance reference. An alternative based on nested cross-validation, with a joint feature–model search is presented in Section 3.

2.1. Problem formulation and evaluation

The training set (X, y) comprises $X \in \mathbb{R}^{5000\times 500}$ with a roughly balanced binary response. Each customer is ranked by an estimated acceptance probability $\hat{p} = P(y{=}1 \mid x)$; the campaign contacts the top $k \leq 1000$ by decreasing $\hat{p}$. Under the business score introduced in Section 1, a contact is profitable in expectation iff $\hat{p} > 1/3$, since $E[\text{contact}] = 10p - 5(1-p) > 0$.

All design choices—feature subsets, model hyperparameters, and the contact count k—are tuned via 5-fold stratified cross-validation. The scorer evaluates held-out fold probabilities at each $k = 1, \ldots, 1000$ and retains the maximiser; out-of-fold (OOF) aggregation avoids in-sample optimism. Models are subsequently refit on the full training set prior to test-time ranking.

2.2. Feature selection

Feature selection proceeds in three stages and yields a ranked shortlist together with prefix subsets `top_01`–`top_26` used throughout the modelling pipeline.

In *Stage 1* (500 → 80 features), near-constant columns are removed by a variance threshold, then mutual information and LightGBM split-gain rankings are combined; the 80 columns with the lowest summed rank are retained. *Stage 2* (80 → 26) removes redundancy via correlation pruning ($|r| \geq 0.85$, discarding the lower-MI member of each pair) and iterative variance-inflation-factor filtering (threshold 5), followed by an ℓ_1-regularised logistic path; the penalty parameter C is chosen by cross-validated business score subject to a sparsity band of 20–30 non-zero coefficients.

Stage 3 orders the surviving features by *drop-column cross-validation*. For each feature f, a classifier is evaluated on all Stage-2 columns except f; the quantity $\Delta_f = \mathrm{score}_{\setminus f} - \mathrm{score}_{\mathrm{full}}$ quantifies the marginal profit loss attributable to f (smaller Δ_f indicates higher importance). Prefix subsets of the resulting ordering are rescored under the same metric. Alternative importance measures (mean decrease in impurity, permutation importance, Boruta) are reported for comparison but are not used to define the primary feature order.

2.3. Baselines

Reference models provide lower bounds on performance and verify the implementation of the business scorer; they do not enter the submission pipeline.

Dummy classifiers are fit on a zero-dimensional feature matrix ($n_{\mathrm{features}} = 0$). The `prior` and `uniform` strategies produce constant acceptance probabilities; the `stratified` strategy yields row-wise random one-hot pseudo-probabilities. A `RandomScoreClassifier` assigns independent $\mathrm{Uniform}(0, 1)$ scores across five random seeds. Learned references comprise logistic regression on the highest-correlation univariate predictor (V390) and on the full 500-dimensional input, the latter illustrating the effect of the feature penalty. Supplementary experiments compare prefix subsets of ranked raw features with an equal number of PCA components, evaluated by top-k F1 and ROC-AUC without the variable-cost term.

2.4. Top-k profit curve

The primary modelling strategy selects a single classifier and feature subset by maximising out-of-fold profit. Logistic regression, LightGBM, and XGBoost are compared on each prefix subset under the cross-validated business score; hyperparameters are tuned on the winning configuration using the same objective rather than ROC-AUC. OOF acceptance probabilities are obtained by 5-fold stratified cross-validation, and the contact count is set to $k^* = \arg\max_k \text{Score}_k$ on the resulting profit curve.

2.5. Expected-value profit targeting

This approach retains the model-selection procedure of Section 2.4 but determines k from economically motivated rules when the profit curve is flat near the contact cap. OOF probabilities are mapped through isotonic regression prior to decision-making. Four candidate values of k are derived: (i) a threshold walk that includes customers while $\hat{p} \geq 1/3$; (ii) the smallest k attaining at least 95% of the peak OOF score; (iii) the maximiser of mean fold-wise business score over a discrete grid of k; and (iv) the unconstrained profit-curve maximum. By default, the *combined* rule $k = \min(k_{\text{ev}}, k_{\text{elbow}}, k_{\text{nested}})$ is adopted.

2.6. Rank fusion committee

We examine whether a committee of heterogeneous rankers, combined by OOF-weighted averaging, improves upon any single expert. The committee comprises four fixed members: logistic regression on `top_03`, LightGBM on `top_05`, logistic regression on `top_08`, and a Borda ranker on `top_05` that aggregates feature-wise signed ranks into pseudo-probabilities. Expert weights are proportional to non-negative cross-validated business scores, $w_e \propto \max(0, \text{CV business}_e)$, and the fused prediction is $\hat{p} = \sum_e w_e \hat{p}_e$. The submission lists the union of all expert feature sets (eight variables).

2.7. Cluster split targeting

Cluster-split targeting introduces customer heterogeneity through hard partitioning before local estimation. For each prefix subset, the number of KMeans clusters $K \in \{2, \ldots, 10\}$ is assessed via inertia and silhouette; we set $K{=}4$ after visual inspection of the diagnostic curves. Within each cross-validation fold, features are standardised, KMeans labels are assigned, and a logistic model is fit per cluster when the training partition contains at least 80 observations; otherwise a fold-level fallback model is used. The same feature set serves both clustering and classification. The optimal prefix subset is selected by comparing OOF business scores across the full sweep.

2.8. Segment-aware targeting

Segment-aware targeting extends the cluster-split framework by separating the feature spaces used for segmentation and for response modelling, and by replacing KMeans with a Gaussian mixture model. In each fold, a GMM with $K{=}5$ components is fit to scaled `top_10` features (optionally after PCA); per-segment logistic regressions are estimated on `top_03` only. Variables used for segmentation are treated as latent and are excluded from the submission, which declares three features. Pooled OOF probabilities are ranked and k is chosen by the profit-curve criterion of Section 2.4.

3. Alternative Approach — Nested CV Methodology

After achieving unsatisfying results in the evaluation of basic experiments, the decision to reorganise and change the policy of testing different setups was made. The main idea was to apply the nested cross-validation methodology alongside multi-stage feature and model selection. The whole procedure was divided into a few research phases so as to ensure the final configuration is stable and robust in terms of the considered business goal. This included: prescreening filtering, inner-CV setup selection, outer evaluation, and final refit. All those stages are described in detail in the further subsections.

3.1. Prescreening filtering

Before fitting any classifier, a prescreening step reduces the initial 500-variable space to a smaller candidate set of k features. Eight ranking methods were considered, spanning three families: filter methods relying on univariate statistics, embedded methods that derive importance from a fitted model, and model-based methods that rank features as part of a more expressive learning procedure. Specifically, mutual information [1] and absolute Pearson correlation [2] serve as filter baselines, while ℓ_1-penalised logistic regression [3] and adaptive best-subset selection [4] serve as embedded selectors. Finally, LightGBM gain importance [5], an Explainable Boosting Machine [6], a sparse GAM fitted via the SPAM coordinate-descent algorithm [7], and a LambdaMART ranker [8] represent model-based methods.

To avoid committing to a single ranking signal, all non-empty subsets of the eight methods were enumerated, resulting in 255 combination strategies. Since each method produces different raw scores, the results are transformed to a normalised rank score in $[0, 1]$:

$$r_i = 1 - \frac{\text{rank}(s_i) - 1}{p - 1},$$

where r_i is the rank score of feature i, $\text{rank}(s_i)$ is its position in the descending ordering, and p is the total number of variables.

Within each multi-method strategy, the individual rank scores are aggregated into a single ensemble score per feature by taking their arithmetic mean:

$$\bar{r}_i = \frac{1}{m} \sum_{j=1}^{m} r_i^{(j)},$$

where m is the number of methods in the strategy and $r_i^{(j)}$ is the rank score of feature i under method j. Features are sorted in descending order of $\bar{r}_i$ to produce the final ranking; for a single-method strategy $\bar{r}_i$ reduces to r_i.

Two additional adaptive strategies that weight methods by their training-set score are also included, bringing the total to 257 prescreen strategies. Each approach was further crossed with a grid of candidate set sizes $k \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20, 30, 40, 50\}$, yielding $257 \times 16 = 4\,112$ (recipe, k) pairs.

3.2. Inner-CV setup selection

The nested CV uses 5 outer folds and 5 inner folds. The outer split contains 4 000 training observations and 1 000 validation points; the inner split uses 3 200 training and 800 validation samples. All prescreening methods are fitted from scratch on each inner training partition, ensuring no label information leaks across folds. This produces a fold-specific feature ranking, from which the 4 112 (recipe, k) pairs are derived. After deduplication of identical feature sets, the top 300 are retained per inner fold — ranked by the mean ensemble score of the selected features — and passed to the model evaluation stage.

The inner-CV selection proceeds in two rounds designed to progressively narrow the search space.

3.2.1. Round 1 - feature-model strategy screening

In Round 1, the 300 feature candidate sets are evaluated using a small set of five screening classifiers (Table 8 in Appendix C). Each model is trained on the inner training partition and evaluated on the inner validation set. The obtained score is scaled by $800/1000 = 0.8$ to be comparable with the outer-fold scale. Those results are averaged across inner folds. The top-60 feature-model strategies per outer fold — ranked by mean inner-fold business score — advance to the next stage.

3.2.2. Round 2 — full model grid

Round 2 evaluates the 60 surviving feature-model strategies against the full model specification grid of 42 configurations, considering seven families: ExtraTrees (8 variants), RandomForest (6 variants), LightGBM (8 variants), XGBoost (8 variants), Elastic-Net logistic regression (6 variants), Explainable Boosting Machine (3 variants), and LambdaMART (3 variants). Each variant explores a different point in the regularisation–capacity trade-off within its family. The tested full hyperparameter grid is listed in Appendix C. The procedure produces 2 580 pipeline candidates per outer fold. The top-20 pipelines per external fold, ranked by mean inner-fold business score, are selected as finalists and passed to the next stage of evaluation.

3.3. Outer evaluation

Rather than selecting finalists independently per outer fold, the 20 candidates for further evaluation are chosen globally - scores from Round 2 are aggregated and the top-20 pipelines by mean inner score across different folds — filtered to at most 6 features — are selected once and evaluated on every external fold.

Each of the 20 candidates is refitted on the outer training partition and scored on the outer validation set. To ensure the evaluation is representative of the final deployment setting, a *rate-matched* strategy is applied. Only the top-200 predictions are scored and the result is scaled by 5 to a 5 000-observation equivalent.

Evaluation is carried out along two parallel tracks. The *per-fold track* yields five comparable scaled scores whose mean and standard deviation characterise configuration consistency across data splits. The *pooled track* concatenates held-out predictions from all 5 outer folds into a single 5 000-observation OOF set and rescores globally, providing a single deployment estimate. The top-10 configurations by mean per-fold score are carried forward to the final refit stage. Five best of them are refitted on the full training set.

Table 1: *Top-ranked features after drop-column CV (Stage 3).*

Rank	Feature
1	V390
2	V309
3	V94
4	V88
5	V255

Table 2: *Stage-3 prefix subsets: 5-fold CV business score (with feature penalty) and ROC-AUC.*

Subset	N_{vars}	CV business	ROC-AUC
top_01	1	**2 301**	0.547
top_03	3	1 891	0.551
top_05	5	1 490	0.590
top_08	8	898	0.583
top_10	10	486	0.617
top_15	15	−484	0.619
top_20	20	−1 472	0.625
top_26	26	−2 684	0.627

4. Results

4.1. Feature selection

Drop-column ranking places V390 first (Table 1). The penalised cross-validated business score is maximised by top_01 (2 301) and decreases monotonically with subset cardinality (top_26: $-2\,684$), whereas ROC-AUC continues to increase (Table 2), indicating a divergence between discrimination and profit under the feature penalty.

4.2. Baselines

Table 3 summarises reference scores under 5-fold cross-validation. Dummy classifiers without declared features score approximately 2 475 at k=1000, consistent with the class prior under a balanced sample. Logistic regression on V390 attains 2 301 and therefore falls below the no-skill floor once the single-feature penalty is applied. With all 500 variables declared, the best-tuned logistic model yields $-97\,493$ despite F1 ≈ 0.668; the corresponding OOF profit curve is shown in Appendix 3.

A supplementary comparison contrasts prefix subsets of Stage-3 ranked raw features with an equal number of PCA components (all 500 columns scaled; 25 components fit once). The same logistic penalty C as the best 500-feature baseline is used. Table 4 reports top-k F1 and ROC-AUC only—no business score or feature penalty, since PCA dimensions are not submission variables. Raw ranked features achieve equal or higher F1 at every k; PCA attains slightly higher ROC-AUC only at k=5 and k=10, but not at k=25. This might indicate that more features introduce noise.

4.3. Top-k profit curve

Among the candidate model families, logistic regression on V390 achieves the highest cross-validated business score (2 301; C=0.001 after hyperparameter tuning). The OOF profit curve peaks at k=997 with score 3 290 (565 TP, 432 FP; Table 6, Appendix 4). Tree-based models and enlarged feature subsets fail to improve the penalised objective.

Table 3: *Reference baseline scores (5-fold CV, with feature penalty).*

Baseline	Features	Business	F1@top-k
Prior / uniform dummy	0	2 475	0.666
Stratified dummy	0	2 474	0.665
Random ranker (mean)	0	2 475	0.665
Logistic (V390)	1	2 301	0.668
Logistic (500 feat.)	500	−97 493	0.668

Table 4: *Top-k raw ranked features vs. PCA components (5-fold CV logistic regression).*

k	F1 raw	F1 PCA	ROC-AUC raw	ROC-AUC PCA
1	**0.668**	0.666	0.555	0.555
3	**0.669**	0.665	0.570	0.584
5	**0.669**	0.665	0.581	0.591
10	**0.669**	0.666	0.594	0.598
15	**0.670**	0.667	0.597	0.599
25	**0.670**	0.667	0.603	0.598

Table 5: *Cluster-split OOF business score by prefix subset (K=4).*

Subset	k	OOF score
`top_05`	998	**3 280**
`top_03`	996	2 880
`top_01`	999	2 800
`top_04`	1 000	2 660
`top_08`	1 000	2 370
`top_10`	999	2 260
`top_26`	996	−895

Table 6: *OOF results by modelling approach (classic pipeline).*

Approach	Feat.	k	TP	FP	OOF score
Segment targeting	3	998	598	400	**3 380**
Top-k profit curve	1	997	565	432	3 290
Cluster split	5	998	618	380	3 280
EV targeting	1	939	536	403	3 145
Rank fusion	8	994	577	417	2 085

4.4. Rank fusion

The weighted committee attains an OOF score of 2 085 at k=994 when eight features are declared. Although true positives increase to 577 relative to the single-expert logistic model, probability averaging degrades ranking at the head of the list; see Appendix 6.

4.5. Cluster split

Table 5 reports the cluster-split sweep at K=4. The best configuration uses `top_05` (OOF 3 280 at k=998; five features). Enlarging the feature set beyond `top_05` reduces OOF performance; diagnostic plots appear in Appendix 7–8.

4.6. Segment-aware targeting

Segment-aware targeting attains the highest OOF score in the classic pipeline—3 380 at k=998 (598 TP, 400 FP) with three declared features (Table 6, Appendix 5). Relative to cluster split, the confusion counts are marginally less favourable, yet the reduction of two submitted variables (−400 penalty) produces a net gain of 100 points.

4.7. Approach comparison

Table 6 and Figure 1 aggregate performance across the five modelling strategies. Segment-aware targeting occupies the first rank; rank fusion occupies the last.

4.8. Nested-CV alternative

All 20 globally-selected candidate pipelines were consistently evaluated across all 5 outer folds. The top-10 by mean rate-matched gross score all use ExtraTrees classifiers, with sparse GAM SPAM (5 configs, k=6) and LightGBM gain (3 configs, k=6) as the dominant prescreeners; the remaining two combine both methods (k=5). Four variables — V255, V380, V176, V342 — appear across all 5 outer folds in every finalist pipeline, confirming a stable and compact signal set.

The winning configuration uses the sparse GAM SPAM prescreener with k=6 variables (V176, V191, V215, V342, V380, V483) and ExtraTrees (depth=5, min_samples_leaf=16, 400 trees, balanced). It achieves a mean per-fold gross score of 5 450 (±130) and a pooled OOF score of 5 425 — a 61% improvement over the best classic-pipeline result of 3 380. The second-ranked configuration (same prescreener, shallower tree) achieves a slightly lower mean (5 400) but substantially lower variance (±64), suggesting that model depth trades mean performance for stability. Table 7 summarises the top-5 configurations. Figure 2 shows business scores across all evaluation scenarios; further diagnostics appear in Appendix B.

Table 7: *Top-5 nested-CV configurations by mean outer gross score.*

Prescreener	Model	k	Mean	Std	OOF
SPAM	ET depth=5, leaf=16, trees=400	6	**5 450**	130	5 425
SPAM	ET depth=5, leaf=20, trees=500	6	5 400	64	5 295
SPAM	ET depth=6, leaf=20, trees=400	6	5 355	151	5 410
SPAM	ET depth=4, leaf=12, trees=320	6	5 320	309	5 395
SPAM	ET depth=3, leaf=8, trees=320	6	5 310	257	5 445

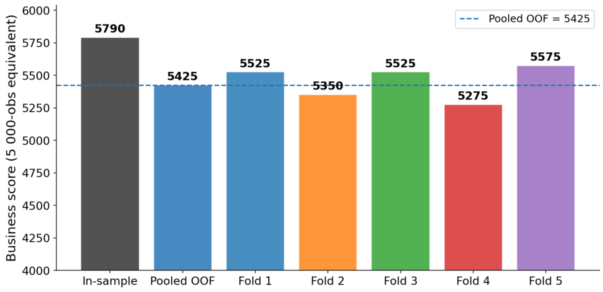

Figure 2: *Business scores across evaluation scenarios. Per-fold scores are rescaled to the 5 000-observation equivalent.*

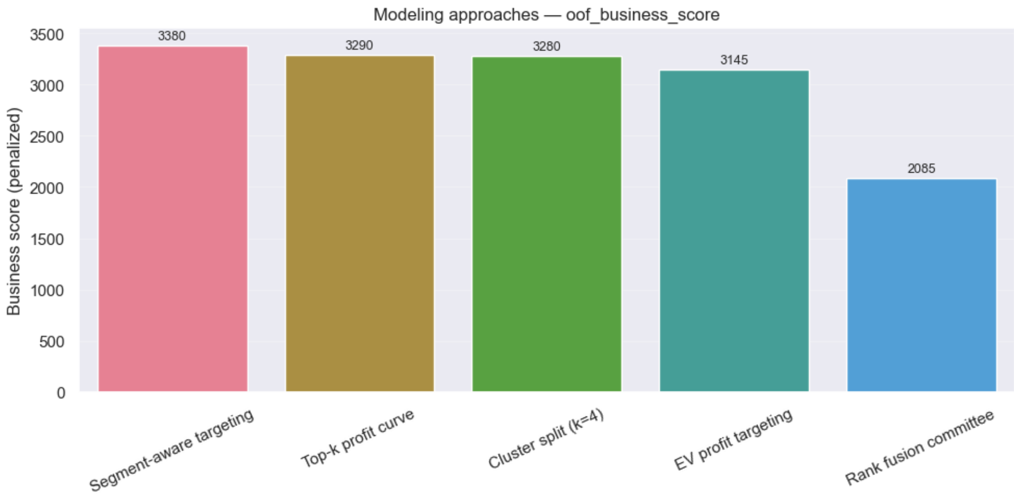

Figure 1: *OOF business score across modelling approaches.*

5. Conclusions

The experiments demonstrate that profit-aligned evaluation substantially changes which features and models are competitive. In the classic pipeline, a single dominant variable (V390) maximises the penalised cross-validated score; additional features reduce the business objective despite improving ROC-AUC, consistent with the high cost of each extra predictor. Segment-aware targeting achieves the best classic-pipeline result (OOF 3 380, three features); rank fusion ranks last despite its higher true-positive count, owing to the eight-feature penalty.

The nested cross-validation alternative lifts the pooled OOF score to 5 425 by enforcing a jointly searching over 257 pre-screening recipes and 42 model configurations. The winning configuration—six features selected by a sparse GAM pre-screener with a shallow ExtraTrees classifier—represents a 61% improvement over the classic pipeline. The low variance of the second-ranked configuration (± 64 vs. ± 130 for the winner) suggests that shallower trees trade marginal mean performance for greater fold-to-fold consistency, which may be preferable in deployment.

6. References

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A. Supplementary figures

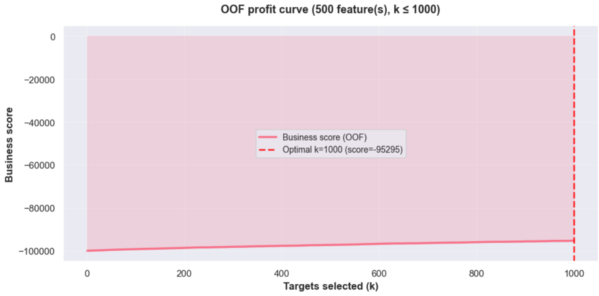

Figure 3: *OOF profit curve for 500-feature logistic regression (TP = 647 at k=1000, score −95 295).*

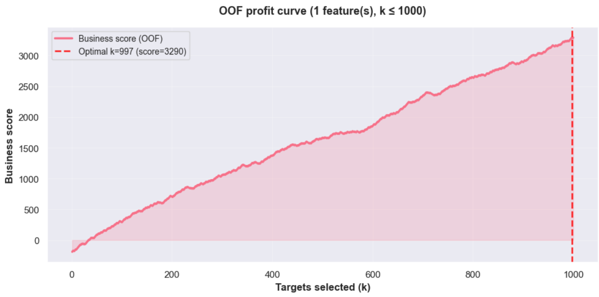

Figure 4: *Top-k profit curve: logistic regression on V390 (k=997, OOF 3 290).*

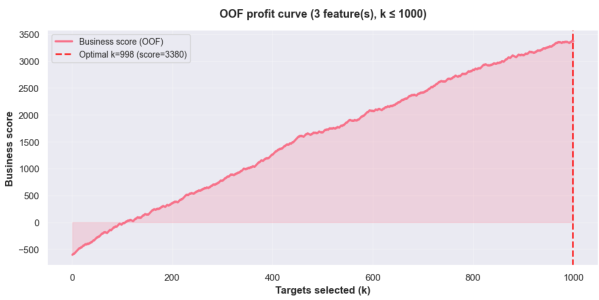

Figure 5: *Segment-aware targeting OOF profit curve (k=998, OOF 3 380).*

B. Nested-CV evaluation diagnostics

Figure 10 shows the distribution of per-fold gross scores across all 20 evaluated configurations; the winning pipeline exhibits both high mean and moderate spread. Figure 11 plots the cumulative false-positive rate through each score band, illustrating how quickly FP accumulates as the selection threshold is lowered. Figure 12 reports precision per score band across all outer folds, confirming that ranking quality is consistent across data splits and degrades gradually toward the contact boundary.

C. Full model specification grid

Table 8 lists the five screening classifiers used in Round 1. Tables 9–11 list all 42 model configurations used in Round 2.

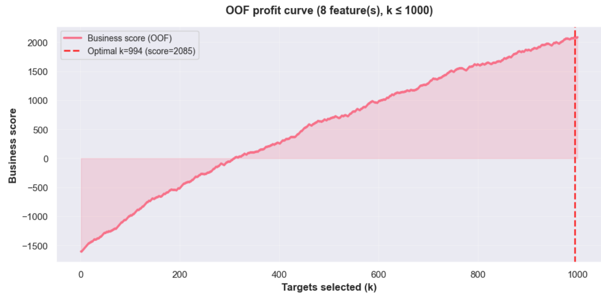

Figure 6: *Rank fusion committee OOF profit curve (OOF 2 085, eight features).*

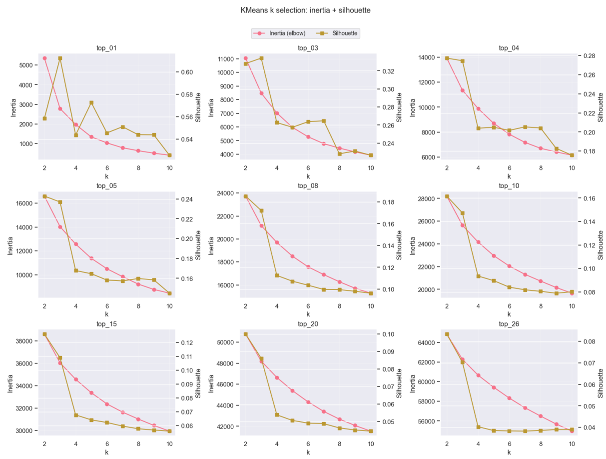

Figure 7: *KMeans diagnostics (inertia and silhouette) for cluster split.*

Table 8: *Round 1 screening model configurations.*

Model	Key hyperparameters	Family
ExtraTrees [9]	depth=4, min_leaf=12, trees=320	Bagging
RandomForest [10]	depth=5, min_leaf=16, trees=400	Bagging
LightGBM [5]	lr=0.035, leaves=15, trees=260	Boosting
XGBoost [11]	lr=0.035, depth=3, trees=260	Boosting
Logistic Elastic Net [12]	C=0.2, ℓ_1-ratio=0.5, balanced	Linear

balanced = `class_weight="balanced"`, *bal_sub* = `class_weight="balanced_subsample"`, *spw* = `scale_pos_weight_multiplier`.

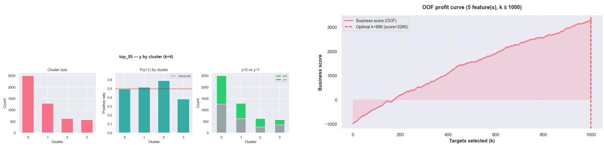

Figure 8: *Cluster split on* `top_05`*: cluster overview (left) and OOF profit curve (right, score 3 280).*

Figure 9: *Cluster split: per-cluster positive-rate profiles.*

Table 9: *ExtraTrees (8 variants) and RandomForest (6 variants).*

Model	depth	leaf	trees	class_weight
ET-1	3	8	320	balanced
ET-2	4	12	320	balanced
ET-3	5	16	400	balanced
ET-4	6	20	400	balanced
ET-5	7	24	500	balanced
ET-6	8	30	500	balanced
ET-7	—	35	400	balanced
ET-8	5	20	500	none
RF-1	4	10	320	bal_sub
RF-2	5	16	400	bal_sub
RF-3	6	20	400	balanced
RF-4	7	24	500	balanced
RF-5	8	30	500	none
RF-6	—	40	600	bal_sub

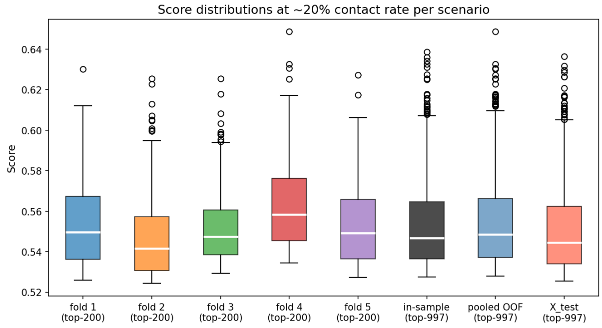

Figure 10: *Score distributions across evaluation scenarios.*

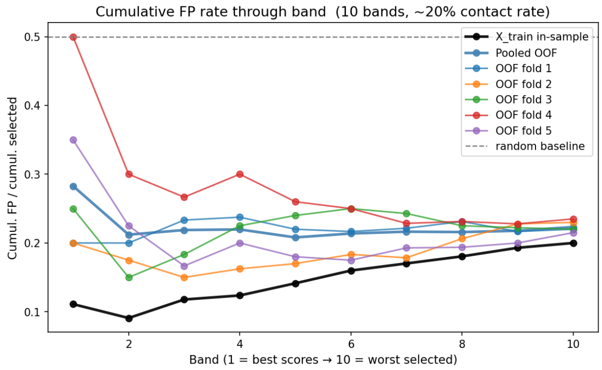

Figure 11: *Cumulative FP rate by score band.*

Table 10: *LightGBM (8 variants) and XGBoost (8 variants). min_child:* `min_child_samples` *for LGB;* `min_child_weight` *for XGB.*

Model	lr	min_child	trees	topology	reg / other
LGB-1	0.050	10	160	7 leaves	spw=0.75
LGB-2	0.040	15	220	11 leaves	spw=0.90
LGB-3	0.035	20	260	15 leaves	spw=1.00
LGB-4	0.030	25	300	23 leaves	spw=1.10
LGB-5	0.025	30	360	31 leaves	spw=1.25
LGB-6	0.020	35	420	39 leaves	spw=1.40
LGB-7	0.040	40	260	15 leaves	col/sub=0.8
LGB-8	0.030	50	320	23 leaves	λ=3.0
XGB-1	0.050	1.0	160	depth=2	spw=0.75
XGB-2	0.040	2.0	220	depth=2	spw=1.00
XGB-3	0.035	3.0	260	depth=3	spw=1.00
XGB-4	0.030	5.0	300	depth=3	spw=1.20
XGB-5	0.025	5.0	360	depth=4	spw=1.30
XGB-6	0.020	8.0	420	depth=4	spw=1.50
XGB-7	0.030	8.0	300	depth=2	col/sub=0.8
XGB-8	0.025	10.0	360	depth=3	λ=3.0

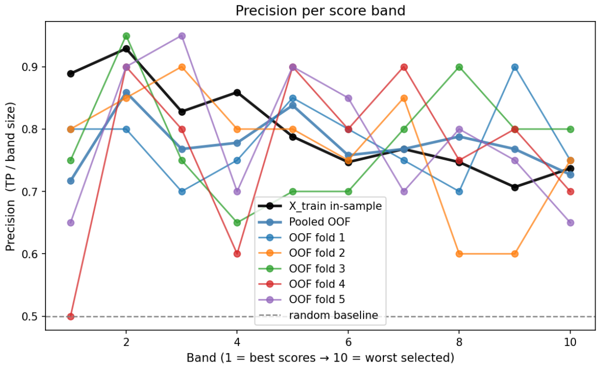

Figure 12: *Precision per score band across outer folds.*

Table 11: *Logistic Elastic Net (6 variants), EBM (3 variants), LambdaMART (3 variants).*

Model	Param 1	Param 2	Param 3
LR-1	C=0.05	ℓ_1=0.0	balanced
LR-2	C=0.10	ℓ_1=0.2	balanced
LR-3	C=0.20	ℓ_1=0.5	balanced
LR-4	C=0.50	ℓ_1=0.8	balanced
LR-5	C=1.00	ℓ_1=1.0	balanced
LR-6	C=2.00	ℓ_1=0.5	none
EBM-1	interact=0	bins=64	rounds=3000, bags=4
EBM-2	interact=0	bins=128	rounds=4000, bags=6
EBM-3	interact=2	bins=64	rounds=3000, bags=4
LM-1	lr=0.05	trees=160	leaves=15
LM-2	lr=0.04	trees=240	leaves=23
LM-3	lr=0.03	trees=320	leaves=31